

VISUALIZATION

Visualizing Tables

Visualizing Tables

- Introduction
- Comparisons
- Compositions
- Correlations
- Distributions
- Geographic Data

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Representing tables

- Two main ways of representing tables
 - Directly (textually): Tables/Spreadsheets
 - Symbolically (pictures): Charts (Information Visualization)
- Use tables when
 - The document will be used to look up individual values
 - The document will be used to compare individual values
 - Precise values are required
 - The quantitative info to be communicated involves more than one unit of measure
- Use charts when
 - The message is contained in the shape of the values
 - The document will be used to reveal relationships among values
- In practice, textual and visual representations are complementary, with effective analysis often involving transitions between the two.

Charts/Graphs

- Visual display that illustrates one or more relationships among entities
- Shorthand way to present information
- Allows a trend, pattern or comparison to be easily comprehended
- Chart Components:
 - Framework: Measurement types, scale
 - Content: Marks, lines, points
 - Labels: Title, axes, ticks

Tasks

- Search
 - Find a known/unknown item
- Query
 - Find out about characteristics of item by itself or relative to others
- Analyze
 - Compare
 - Correlations
 - Compositions
 - Distributions

Basic Plots for Basic Tasks

→ Search

	Target known	Target unknown
Location known	 <i>Lookup</i>	 <i>Browse</i>
Location unknown	 <i>Locate</i>	 <i>Explore</i>

→ Query

→ Identify



→ Compare



→ Summarize



→ ALL DATA

→ Trends



→ Outliers



→ Features



→ ATTRIBUTES

→ One

→ Distribution



↓ Extremes



→ Many

→ Dependency



→ Correlation



→ Similarity



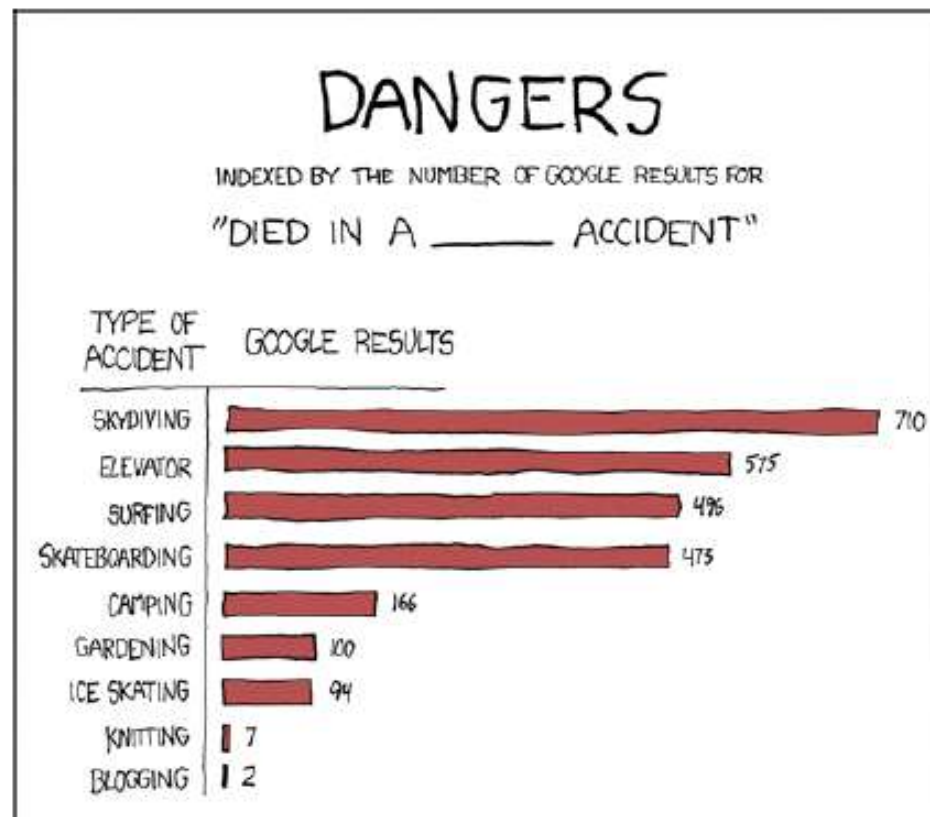
Few's Selection & Design Process

- Determine your message and identify your data
- Determine if a table, or graph, or both is needed to communicate your message
- Determine the best means to encode the values
- Determine where to display each variable
- Determine the best design for the remaining objects
 - Determine the range of the quantitative scale
 - If a legend is required, determine where to place it
 - Determine the best location for the quantitative scale
 - Determine if grid lines are required
 - Determine what descriptive text is needed
- Determine if particular data should be featured and how

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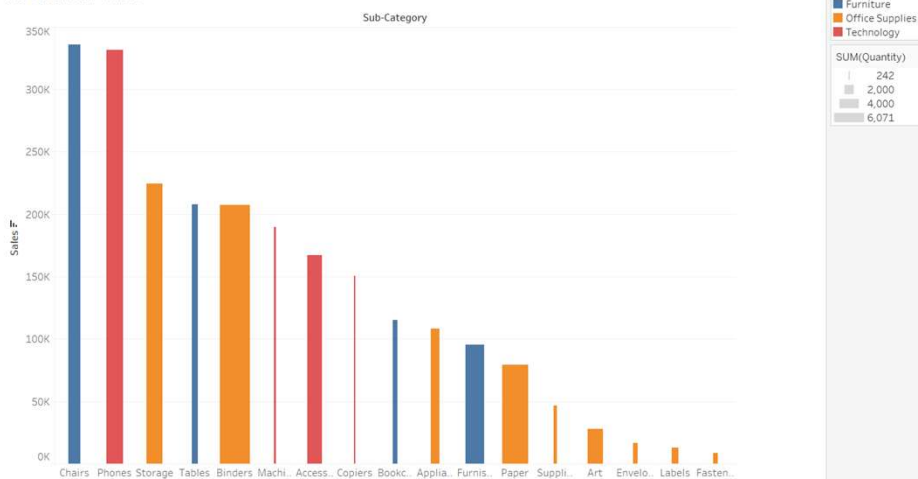
Bar Charts



Good for showing comparisons

Horizontal vs Vertical

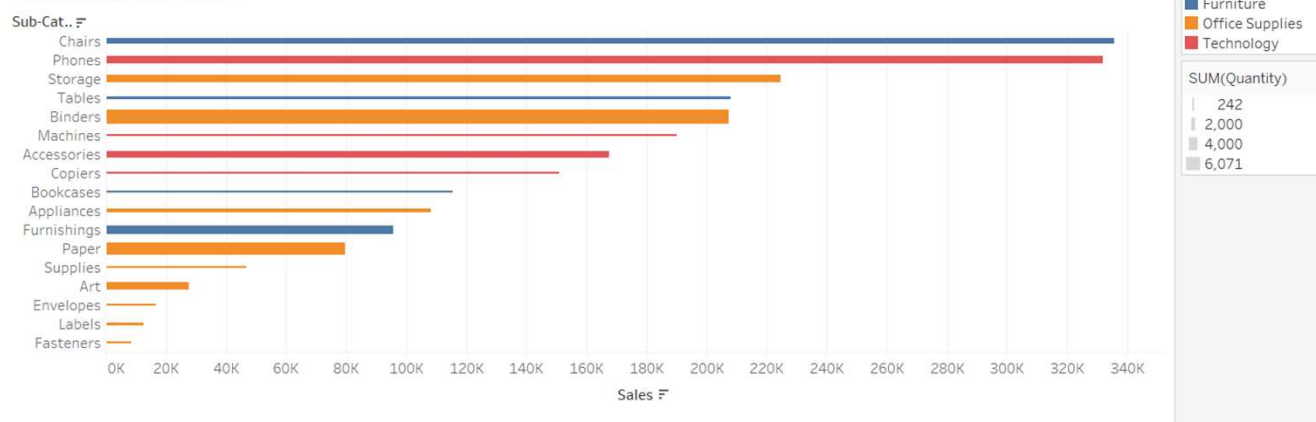
Vertical Bar Chart



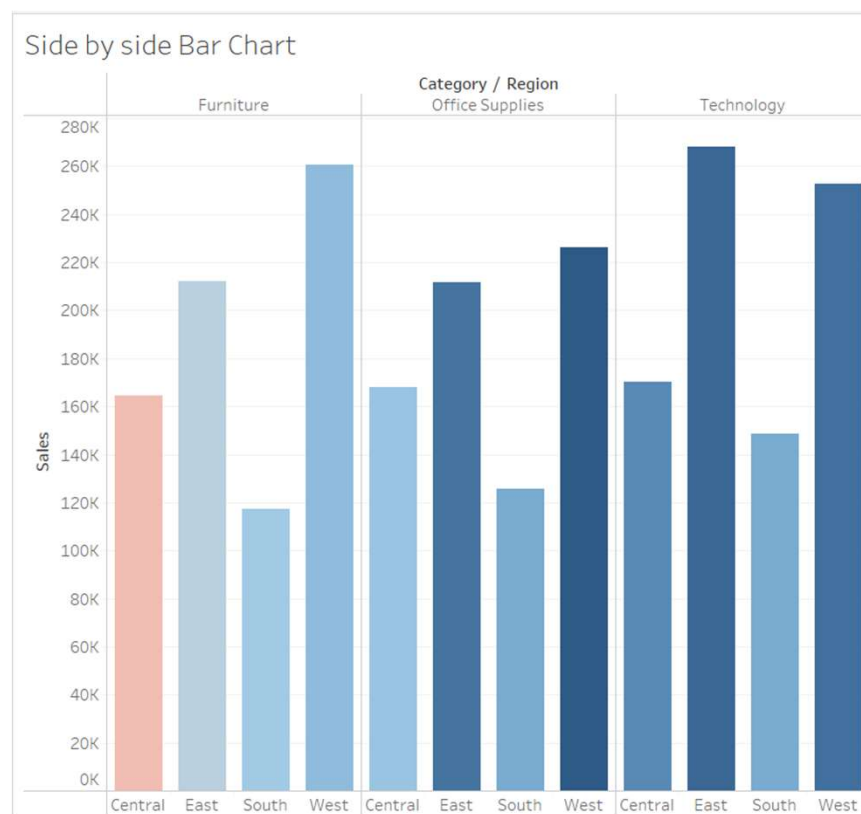
While the vertical bar chart is usually the default, it's a good idea to use a horizontal bar chart when there are long category labels.

In a vertical chart, these labels might overlap and would need to be rotated or shifted to remain legible; the horizontal orientation avoids this issue.

Horizontal Bar Chart



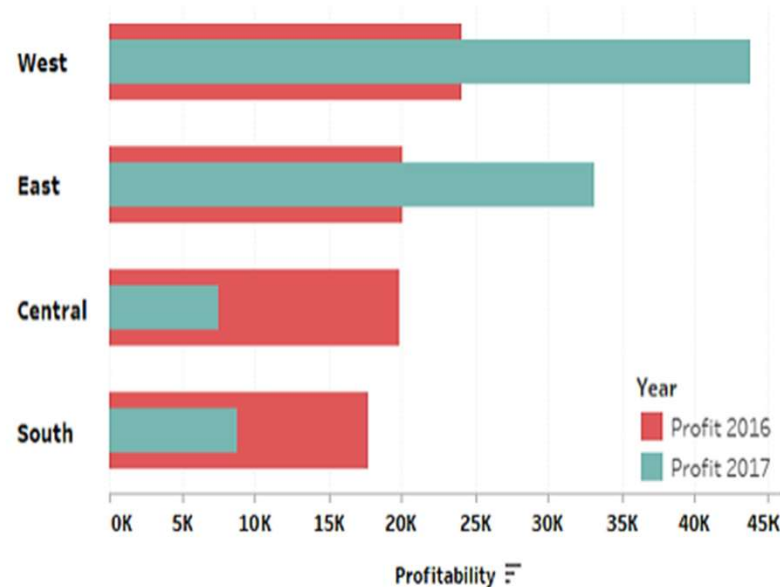
Side-by-side/Clustered Bar chart



Bar in bar Chart

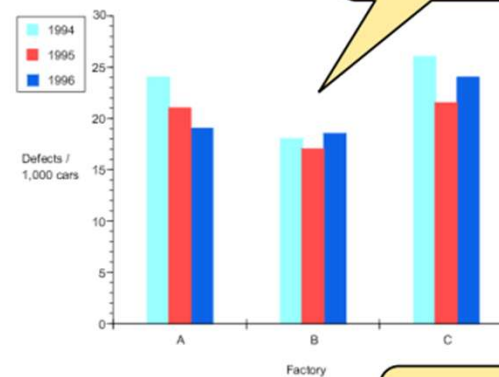
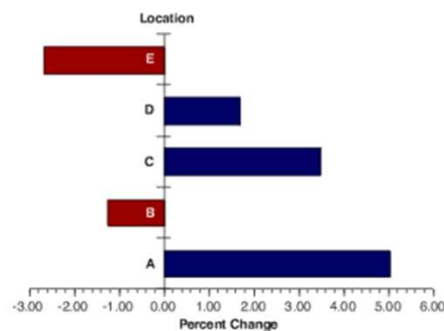
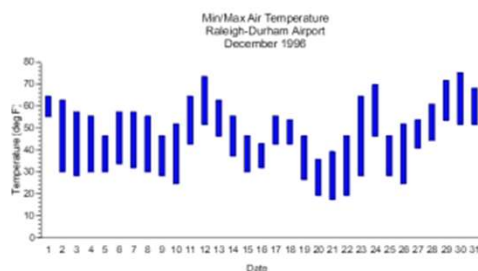
Bar-in-bar chart is simply two interlocking bar charts. Its useful when you need to illustrate change between two points. E.g. Profit between two timelines for different products or regions.

Example of a bar-in-bar chart

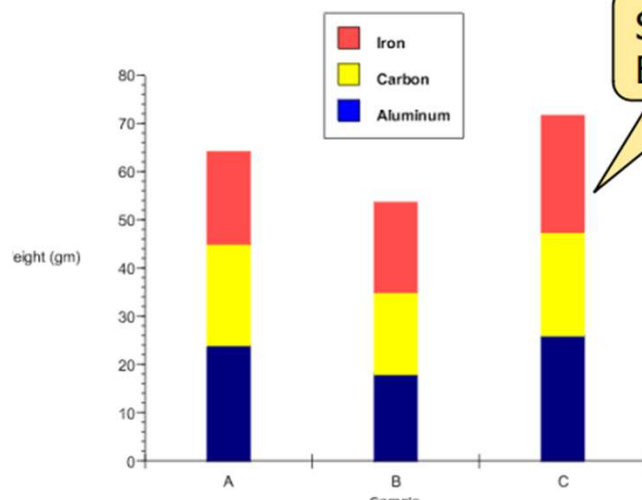


The same information can also be shown as a Side-by-side bar chart

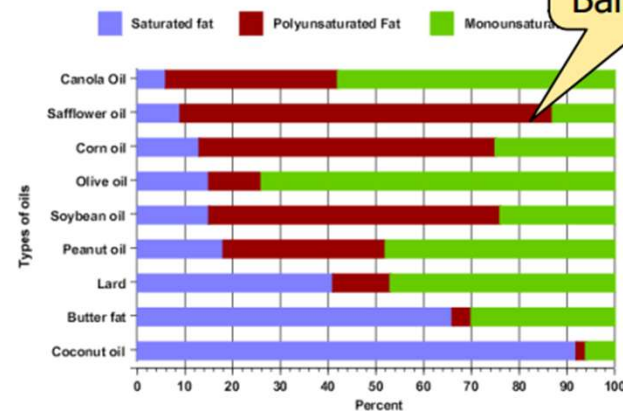
Gallery of Bar Charts



Clustered Bar Chart



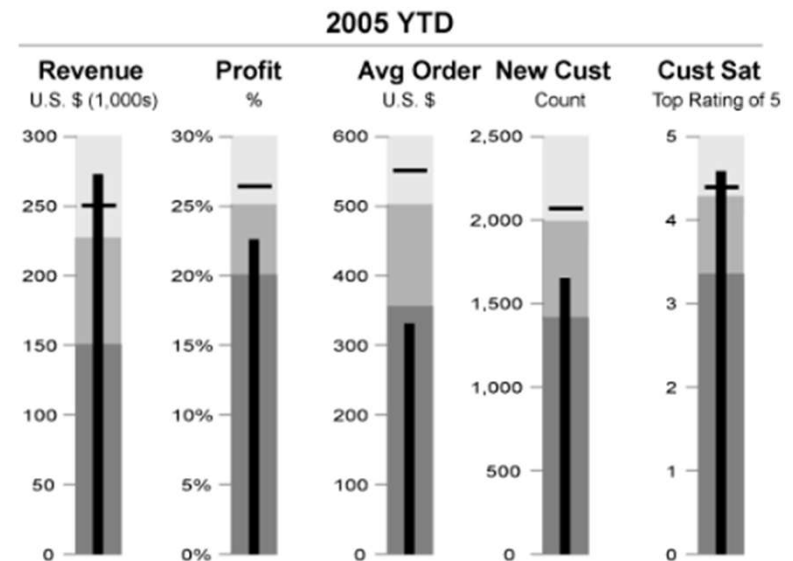
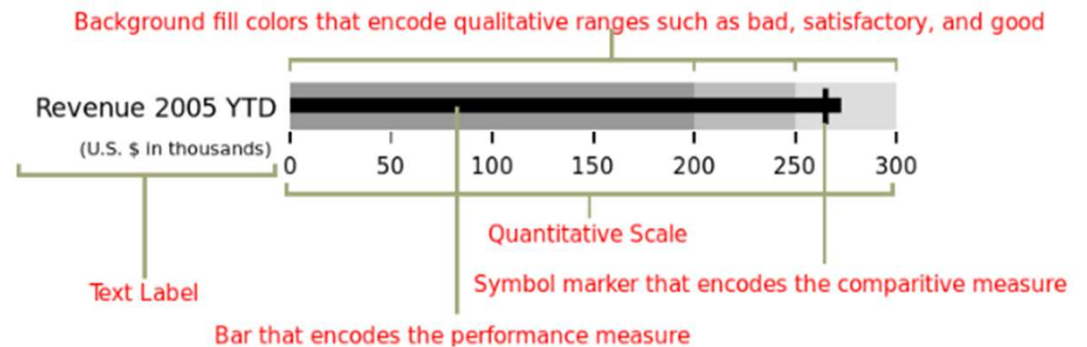
Stacked Bar Chart



Stacked Bar Chart

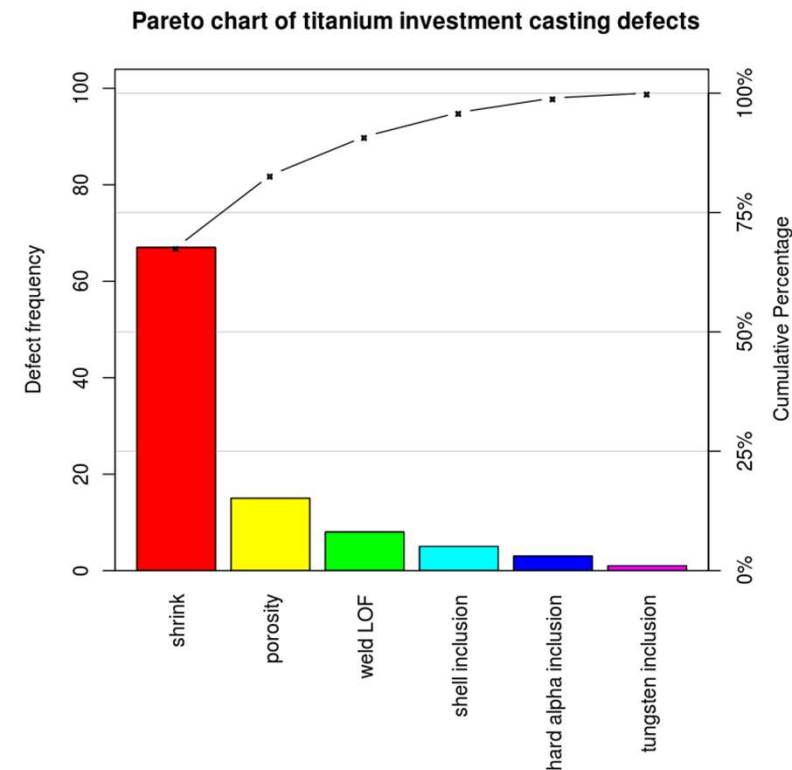
Bullet Graph

- A **bullet graph** is a variation of a bar graph developed by Stephen Few.
- The bullet graph features:
 - A single, primary measure,
 - Compares that measure to one or more other measures to enrich its meaning (for example, compared to a target)
 - Displays it in the context of qualitative ranges of performance, such as poor, satisfactory, and good.
 - The qualitative ranges are displayed as varying intensities of a single hue
 - Bullet graphs may be horizontal or vertical and may be stacked to allow comparisons of several measures at once
- Bullet graphs are particularly useful in dashboard-style summaries and performance monitoring applications where space efficiency is important.

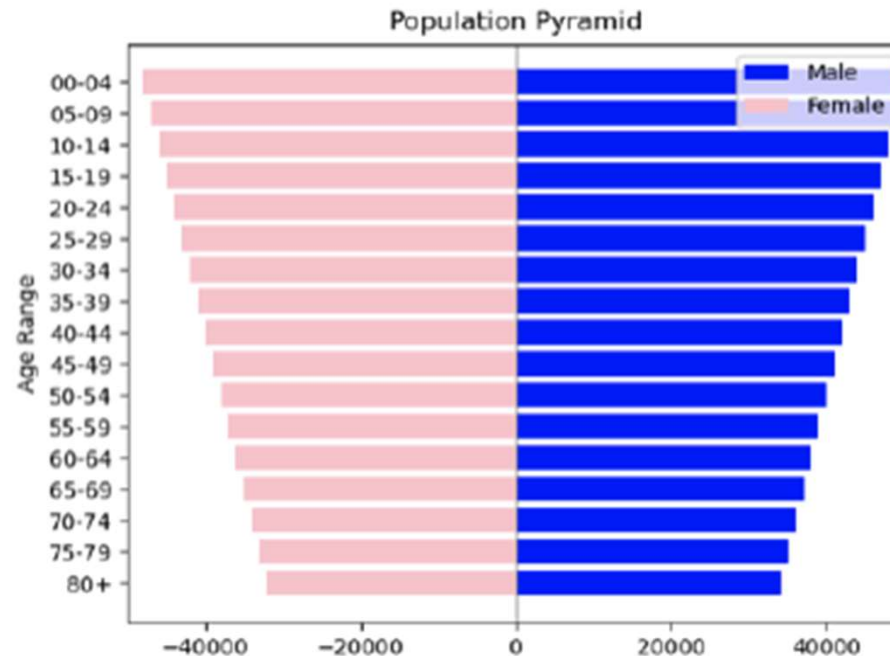


Pareto Chart

- A **Pareto chart** is a type of chart that contains both [bars](#) and a [line graph](#), where individual values are represented in descending order by bars, and the cumulative total is represented by the line.
- The left vertical axis is the [frequency of occurrence](#)
- The right vertical axis is the cumulative percentage of the total number of occurrences, or total of the particular unit of measure.
- The purpose of the Pareto chart is to highlight the most important among a (typically large) set of factors. In [quality control](#) Pareto charts are useful to find the defects to prioritize in order to observe the greatest overall improvement.



(Population) Pyramid Charts

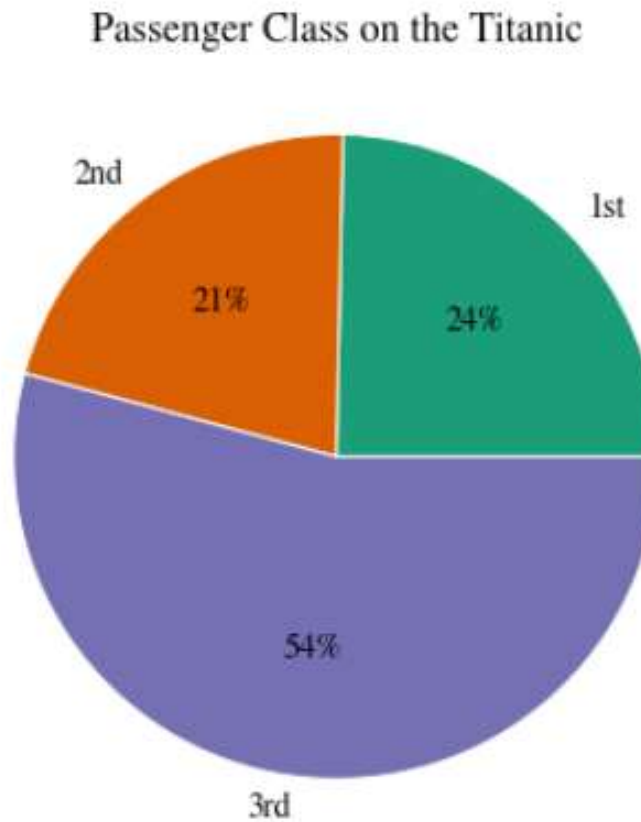


- Population Pyramid charts are used for population data.
- It requires one categorical attribute for ranges (e.g., age groups), another categorical attribute (for example, sex) and a quantitative attribute.

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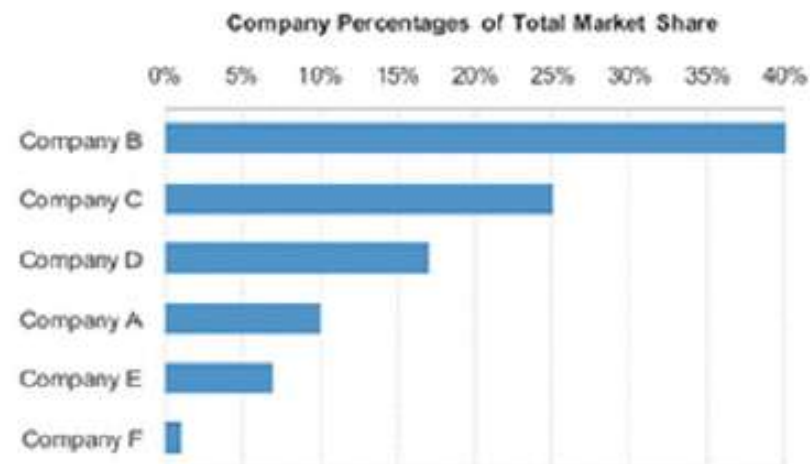
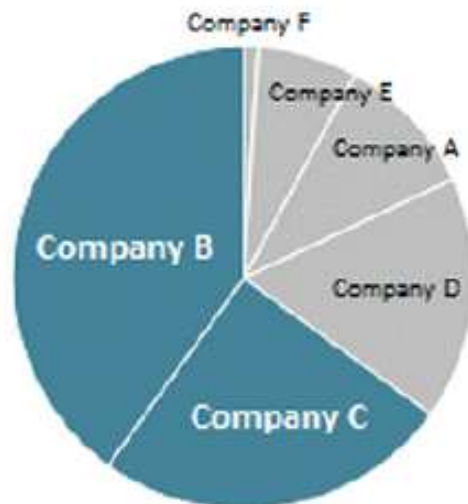
Pie Charts



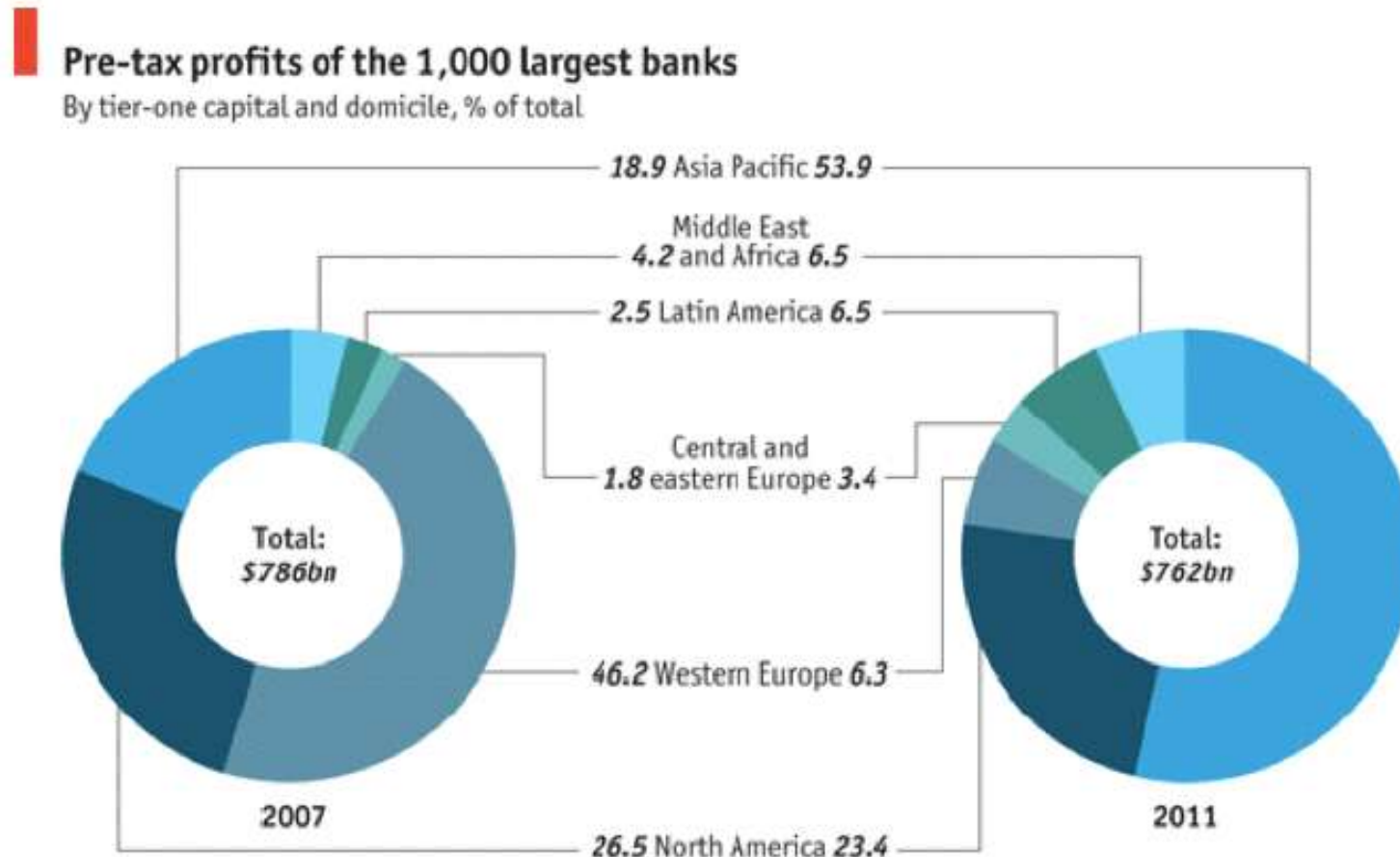
- Pie charts are good for showing compositions.
- However, less effective if:
 - Many categories
 - Small differences between values

Pie vs Bar Charts

65% of the market is controlled by companies B and C



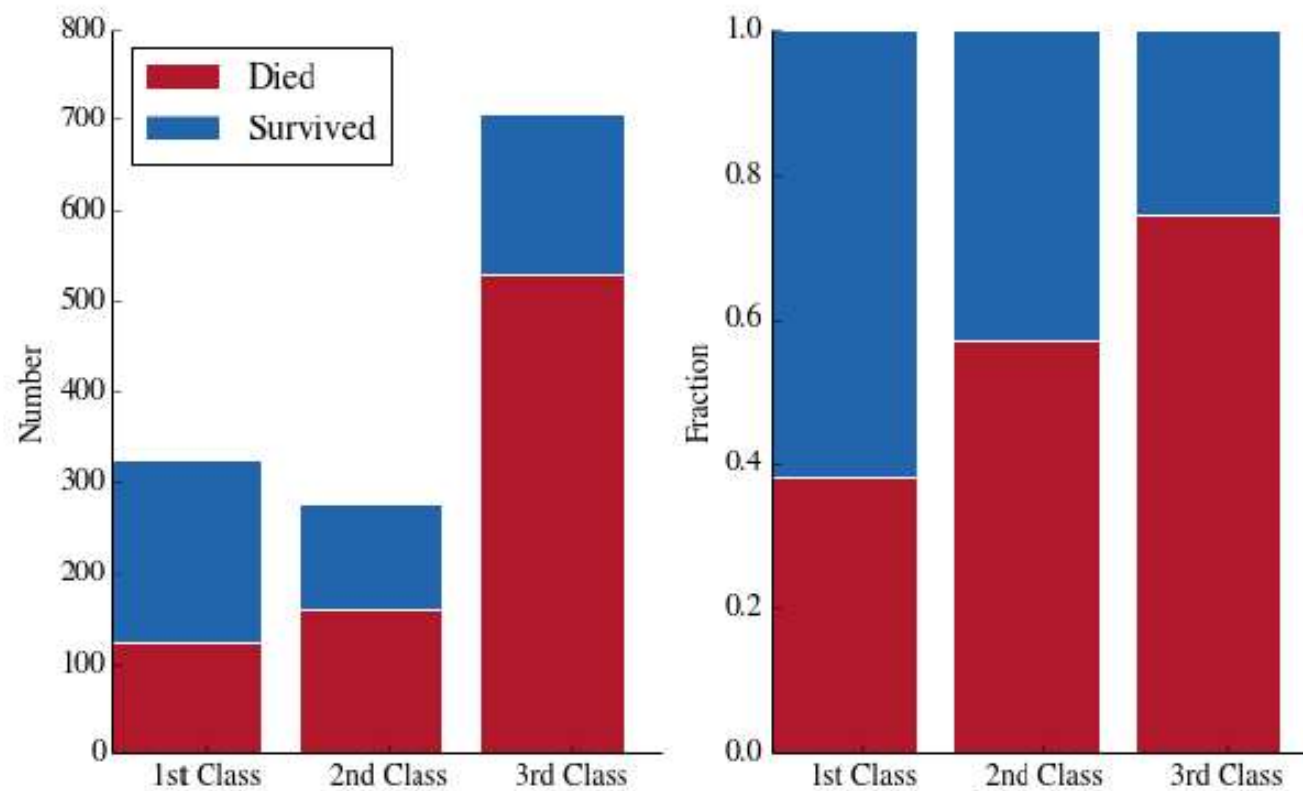
Donut Charts



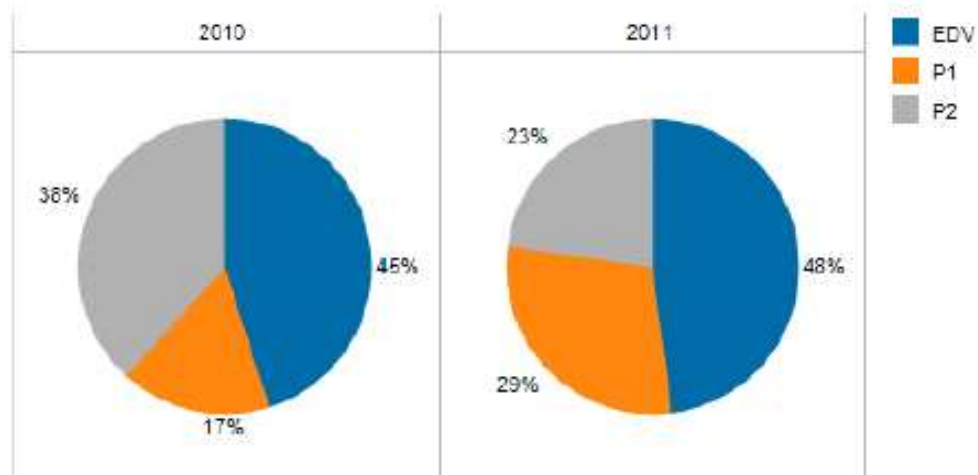
Source: *The Banker Top 1000*

- Because humans are less accurate at comparing angles than lengths, pie and donut charts should be used sparingly and only when the part-to-whole relationship is the primary message.

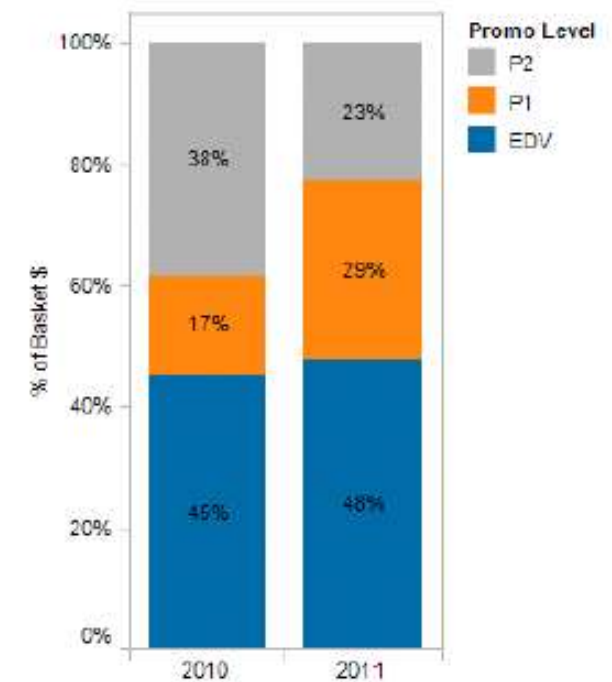
Stacked Bar Charts



Stacked Bar vs Pie Charts

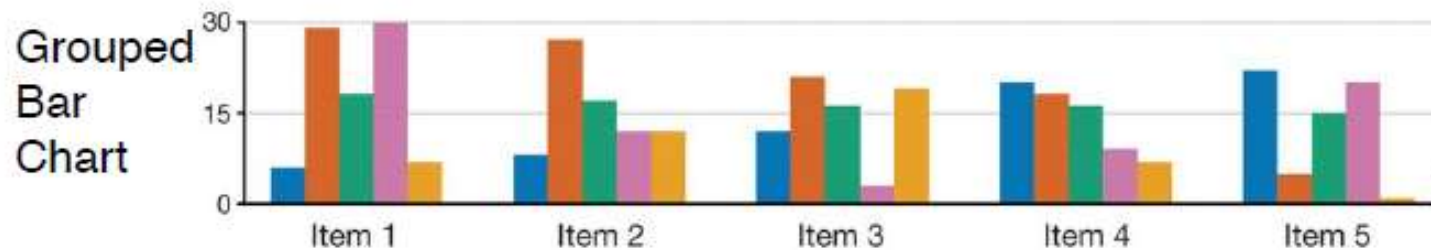
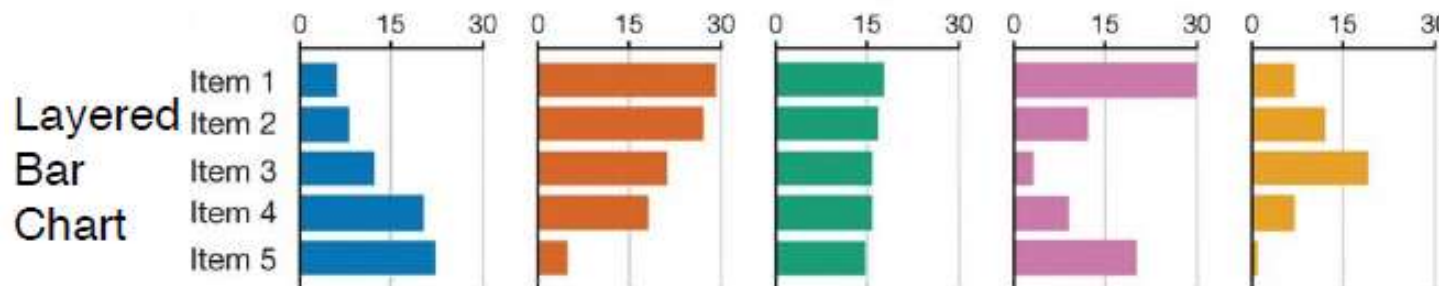
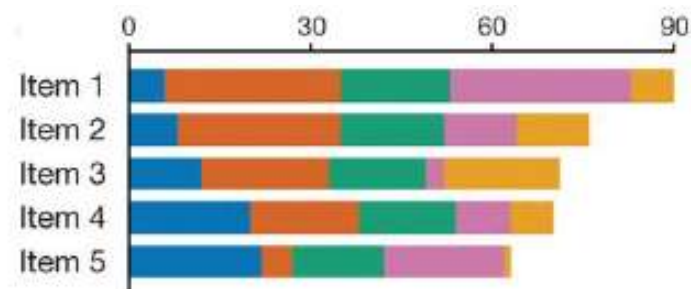


vs.



Comparison of Chart types

Category 1 ●
 Category 2 ●
 Category 3 ●
 Category 4 ●
 Category 5 ●



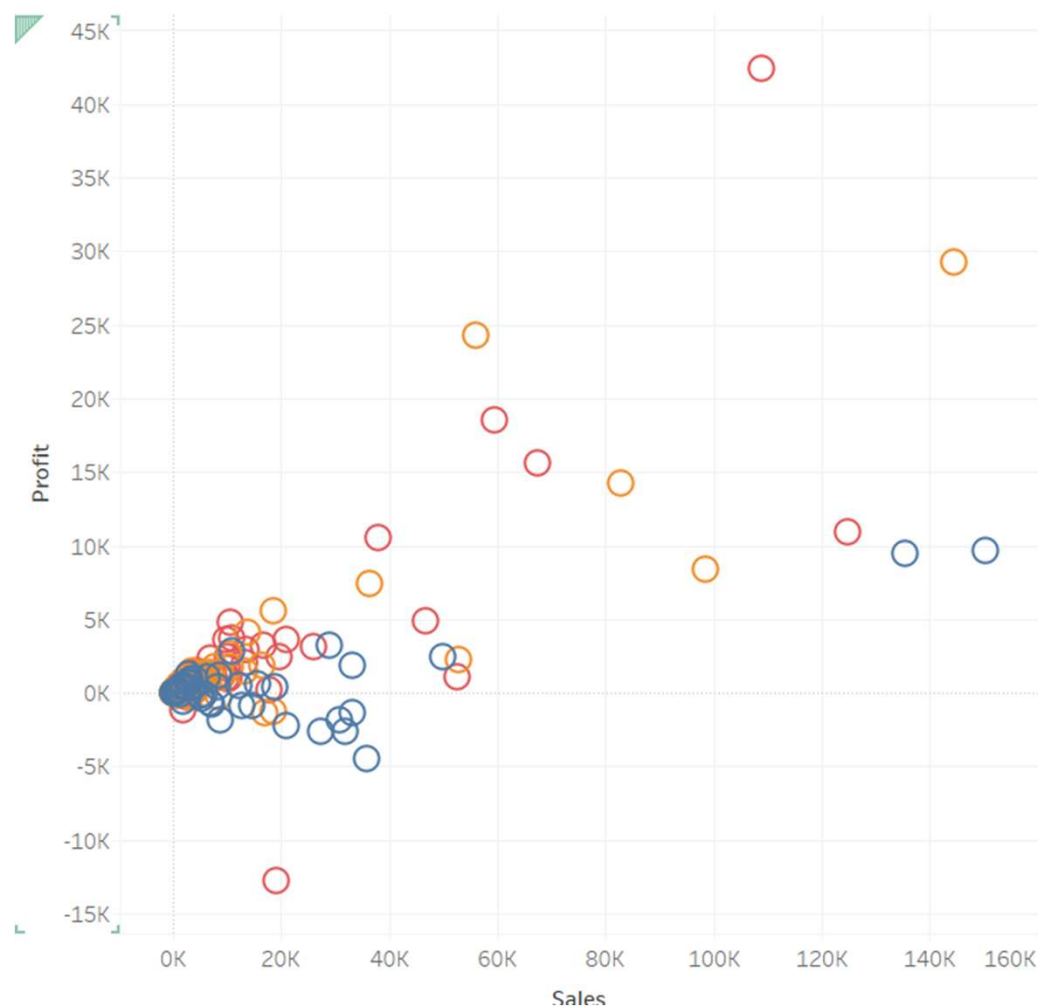
Small Multiples

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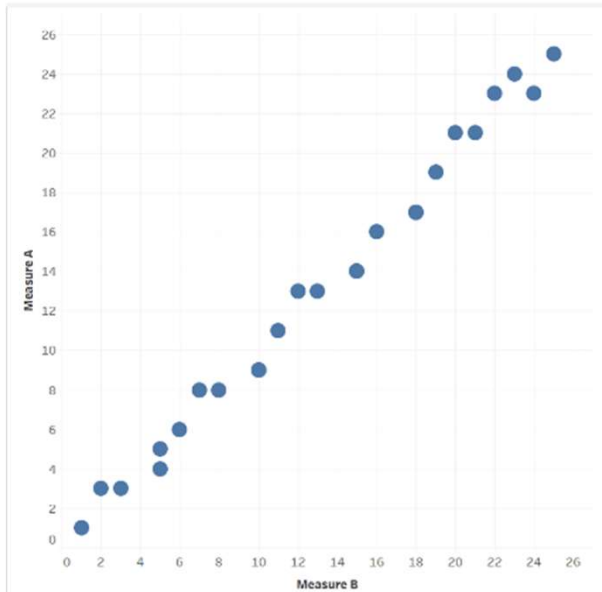
Scatterplots

Scatter plots display two quantitative values on the x and y axis to show the relationship between them.

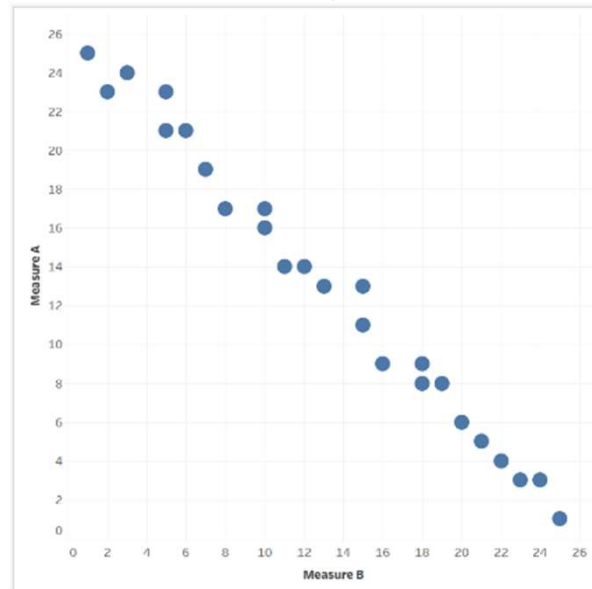


Correlations

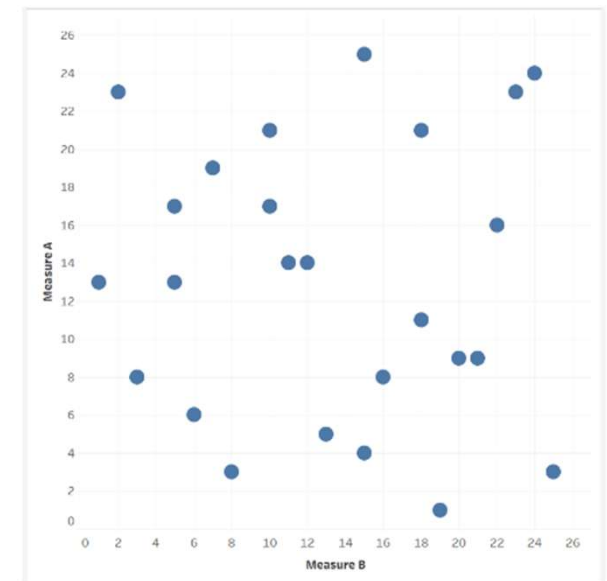
Scatterplots are very useful for showing correlations in 2 dimensional data



(a)



(b)

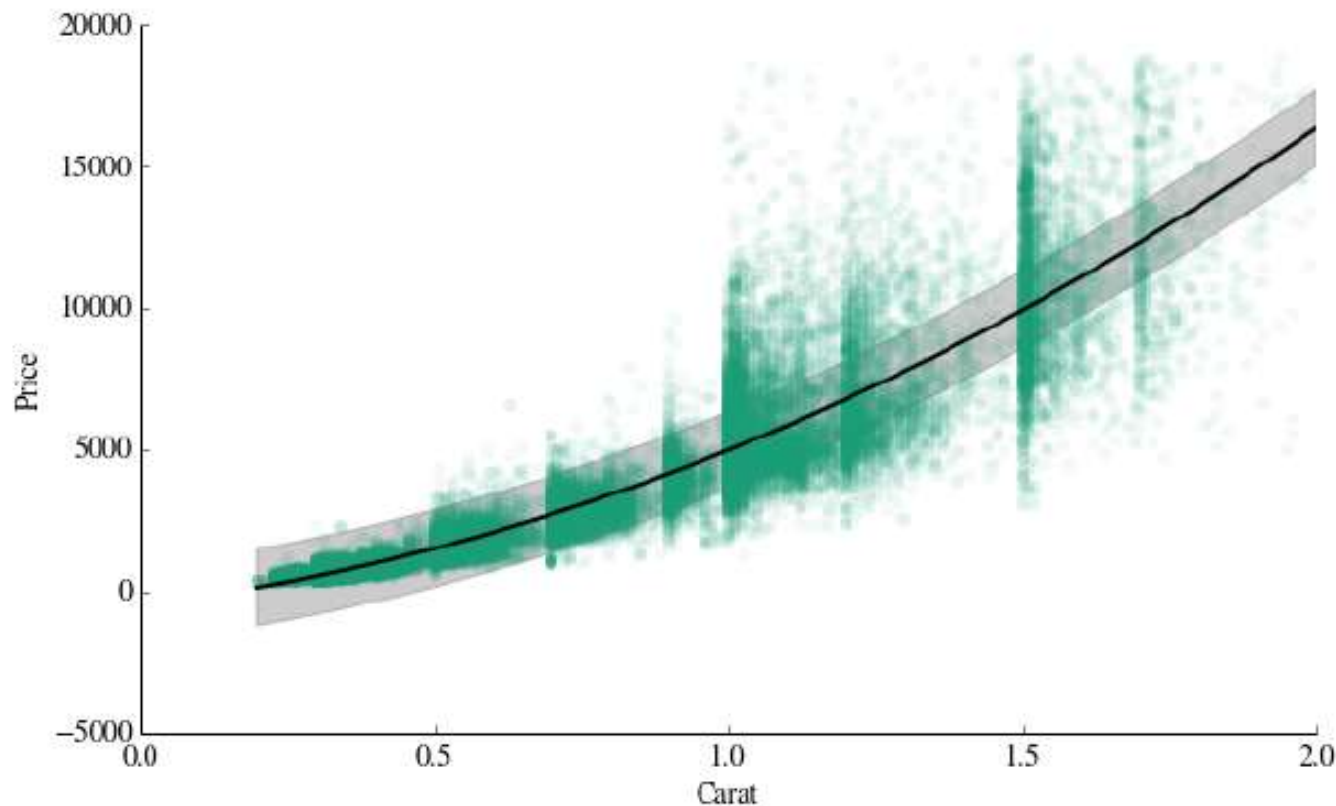


(c)

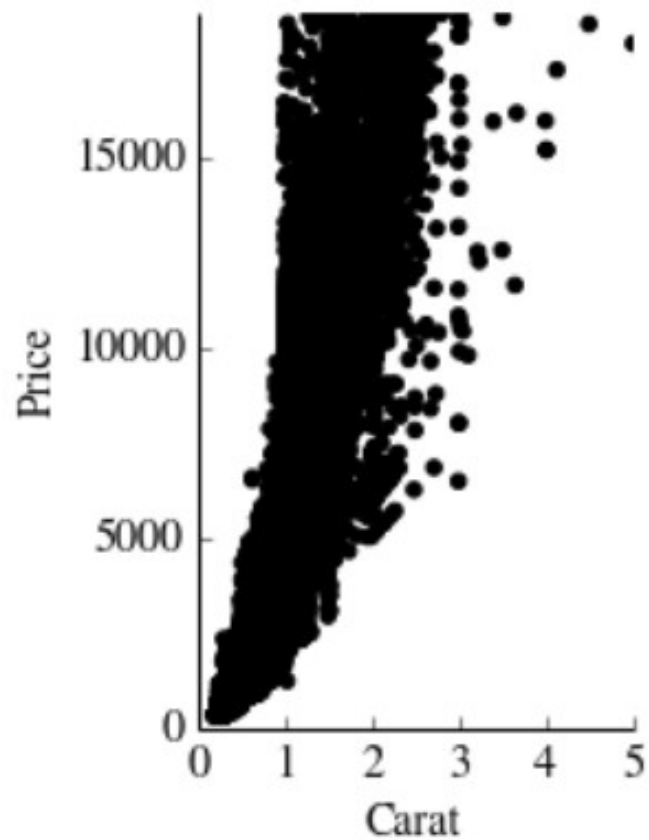
- a) Positive: The values of the variables rise and fall together.
- b) Negative: The values of the variables move in opposite directions.
- c) Zero: There is no relationship between the variables.

Trend Lines

- A trend line in a scatter plot is a line that best fits the data in the plot and shows the pattern of the data.
- Also called the **line of best fit**.
- They can also help identify the relationship between variables.
- The most common type of trend line is a linear trend line, which is a straight line.
- However, a curve may be better for describing some data.



Overplotting

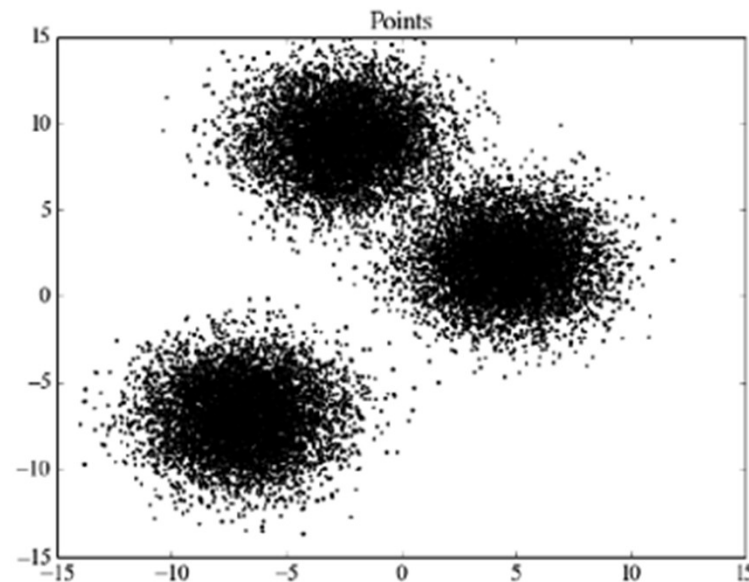


How to deal with overplotting

- Reducing size of data objects
- Removing all fill color from data objects
- Changing the shape of data objects
- Jittering data objects
- Making data objects transparent
- Encoding the density of values
- Reducing the number of values
 - Aggregating the data
 - Filtering the data
 - Breaking the data into a series of separate graphs
 - Statistically sampling the data

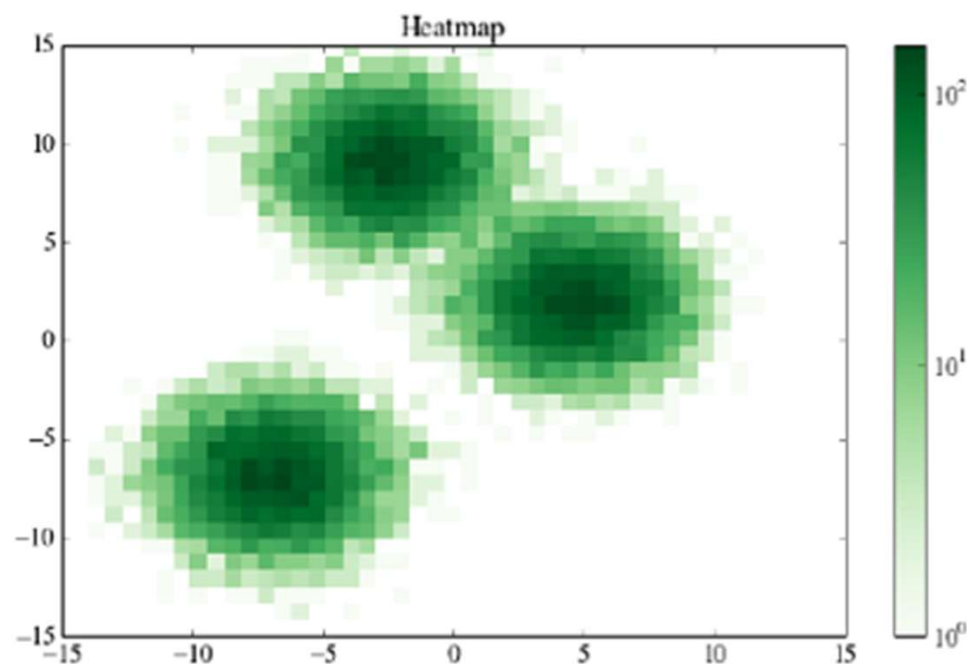
Binned Scatterplots

- **Binned scatterplots** take all data points from the original scatterplot and place each one into exactly one group called a bin.
- Once every observation is in a bin, each bin will get one point on a scatterplot, reducing the amount of clutter on the plot



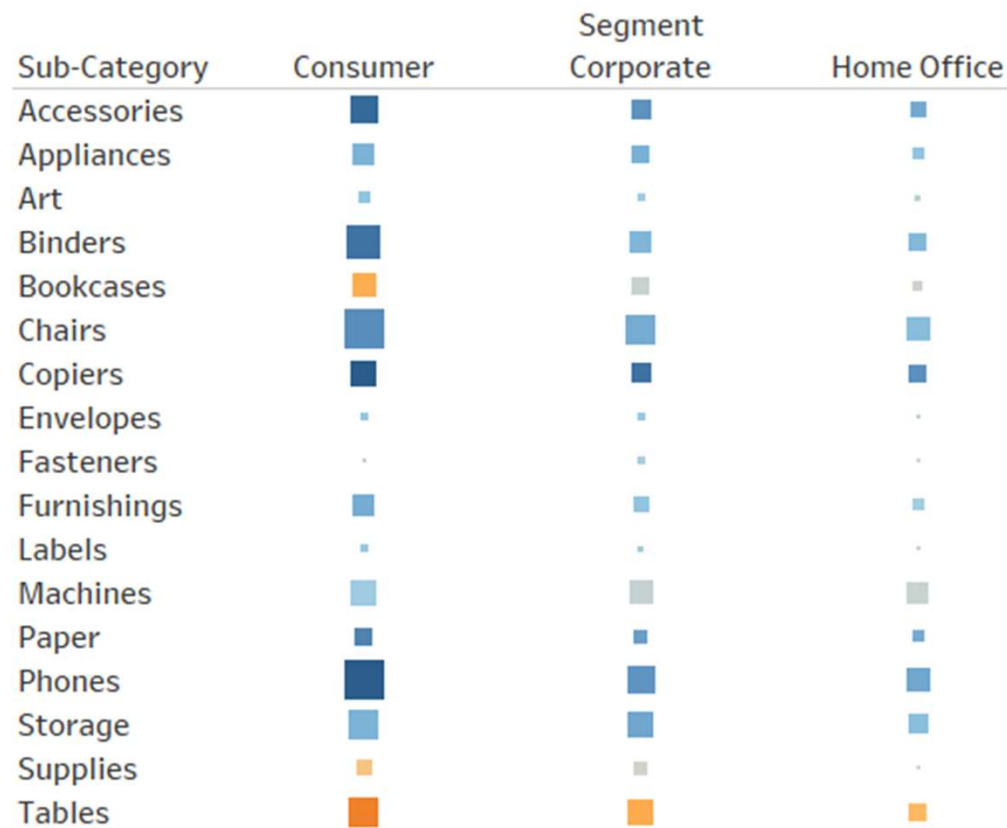
Heat Maps

A heat map is a **two-dimensional representation of data** which uses color to represent the density of the datapoints



Grid Heat Map

- A **grid heat map** displays magnitude as color in a two-dimensional matrix, with each dimension representing a category of trait and the color representing the magnitude of some measurement on the combined traits.

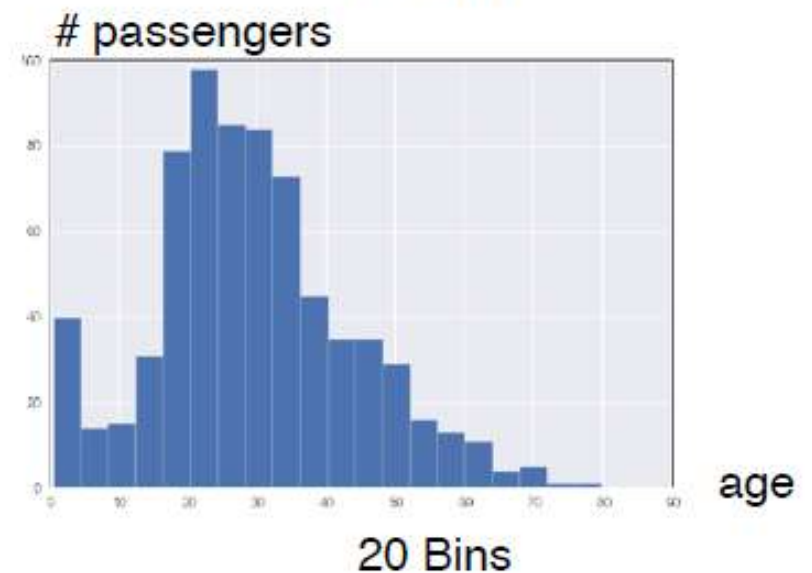
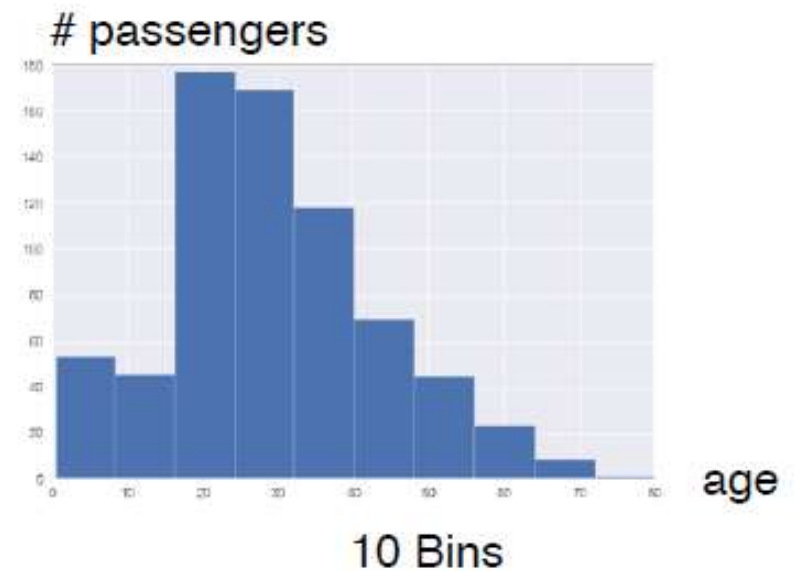


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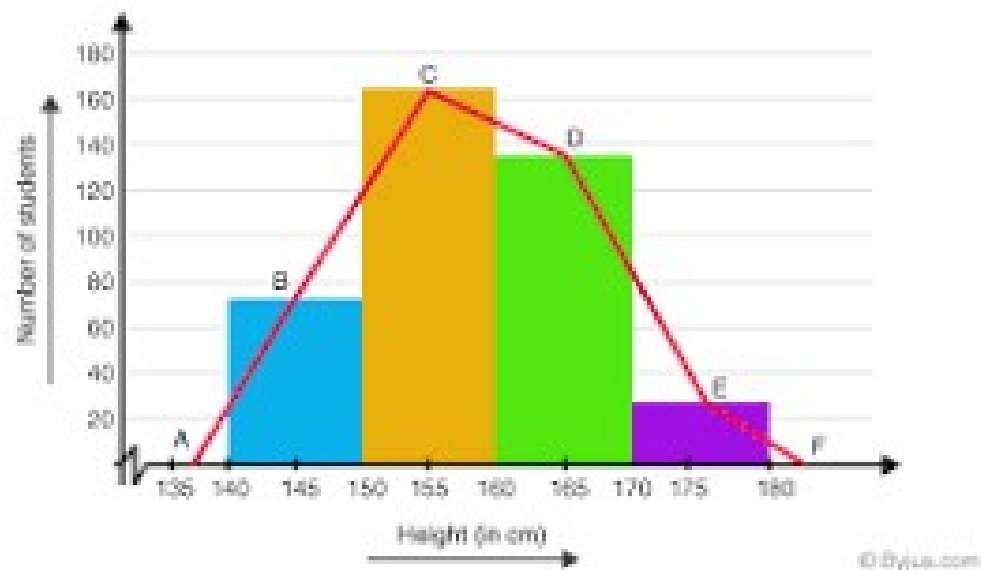
Histograms

- Good for showing distributions
- #bins hard to predict
- rule of thumb: $\#bins = \sqrt{n}$
- Make interactive!



Frequency Polygon

- A frequency polygon is a line graph of class frequency plotted against class midpoint.
- It can be obtained by joining the midpoints of the tops of the rectangles in the histogram

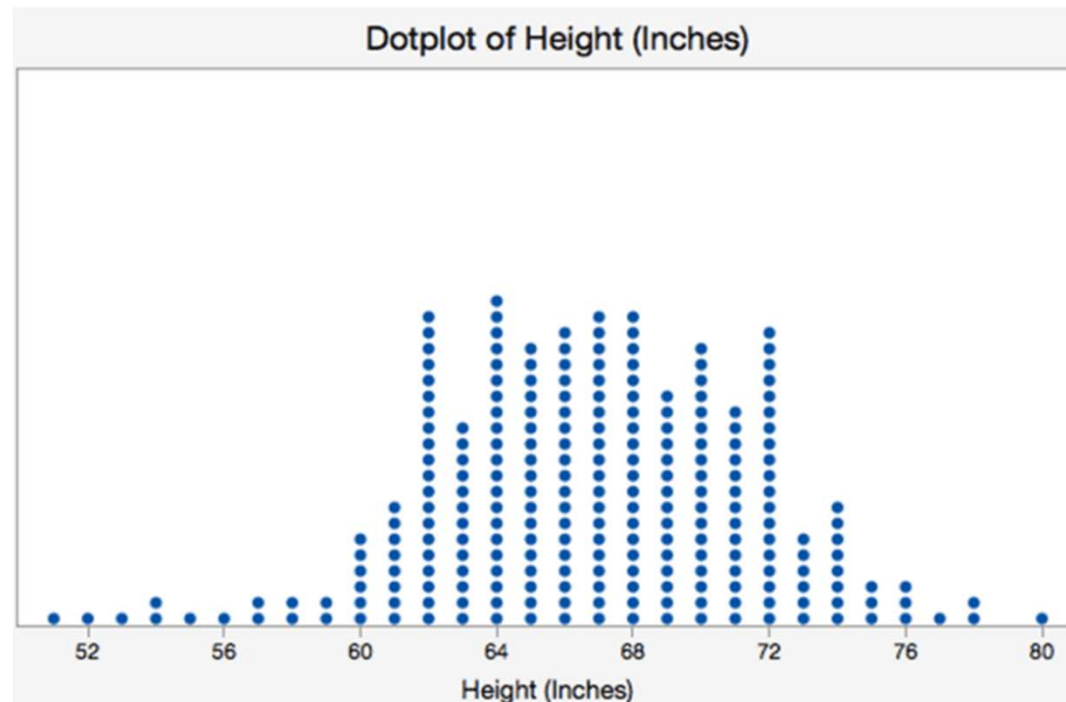


Dot Plots

- A dot plot or dot chart is a simple form of data visualization that consists of data points plotted as dots.
- There are several variations of the dot plot.
- Two most common ones are:
 - Wilkinson dot plot
 - Cleveland dot plot

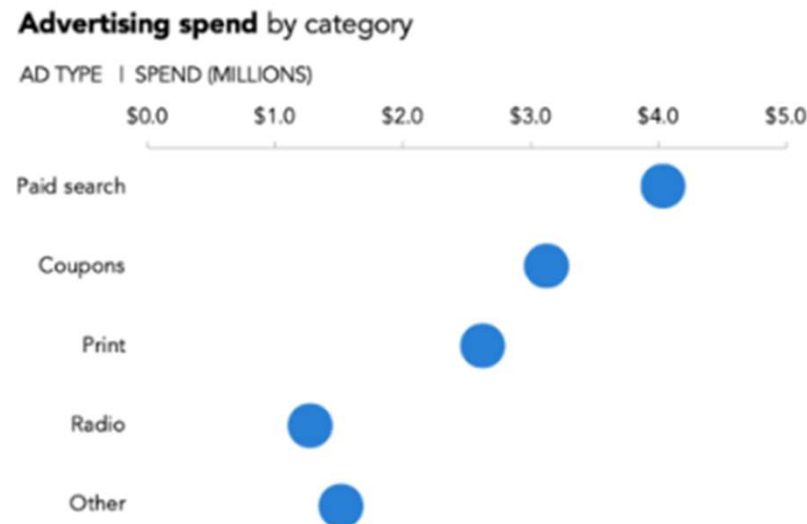
Wilkinson Dot Plot

- Displays a distribution of continuous data, like a histogram, but shows individual data points instead of bin counts.
- Though variations of such plots have been around for more than 100 years, Leland Wilkinson largely standardized the form.



Cleveland Dot Plot

- Developed by William S. Cleveland
- This visualization is used it to compare values across categories.
- Cleveland dot plots are placed horizontally so as to enable easier comparisons among categories.
- They are somewhat like a horizontal bar chart where the bars are replaced by dots at the values associated with each category.



Stem-and-Leaf Plot

- A stem-and-leaf display or stem-and-leaf plot is a device for presenting quantitative data in a graphical format, similar to a histogram, to assist in visualizing the shape of a distribution.
- Unlike histograms, stem-and-leaf displays retain the original data to at least two significant digits, and put the data in order

Stem	Leaf
0	7
1	4
2	1 8
3	5
4	2 9
5	6
6	3
7	0 7
8	4
9	1 8

Multiples of 7

5	10 17 29 39 47 54
6	1 7 13 16 22 26 30 32 36 41 48 48 51 55 58
7	1 4 7 10 13 17 20 22 25 28 31 34 37 40 44 46 50 53 57 59
8	2 5 6 11 14 17 21 24 27 31 35 38 41 44 48 51 54 58
9	2 6 10 14 17 21 24 28 31 35 38 41 44 47 51 55 59
10	3 7 11 16 20 27 30 35 39 41 45 50 54 56
11	0 4 9 11 15 19 24 26 30 34 39 41 45 49 54 56
12	0 4 9 11 15 19 24 26 30 34 39 41 45 49 54 56
13	0 4 9 11 15 19 24 26 30 34 39 41 45 49 54 56
14	0 4 9 11 15 19 24 26 30 34 39 41 45 49 54 56
15	0 4 9 11 15 19 24 26 30 34 39 41 45 49 54 56
16	0 4 9 11 15 19 24 26 30 32 35 39 41 45 47 50 54 57
17	1 5 9 13 16 20 24 28 32 36 40 44 46 50 58 57
18	1 4 8 12 16 19 23 27 31 34 38 42 45 49 58 58
19	1 5 9 13 16 20 23 28 31 35 38 43 45 50 58 58
20	0 5 8 12 16 20 23 27 31 34 38 42 46 49 54 57
21	1 5 8 10 17 21 25 29 32 38 42 46 51 55 59
22	2 7 11 15 19 23 27 31 37 42 47 51 56 59
23	3 9 14 18 22 26 30 34 40 45 48 56
0	0 5 12 16 21 30 34 48

Train time-table

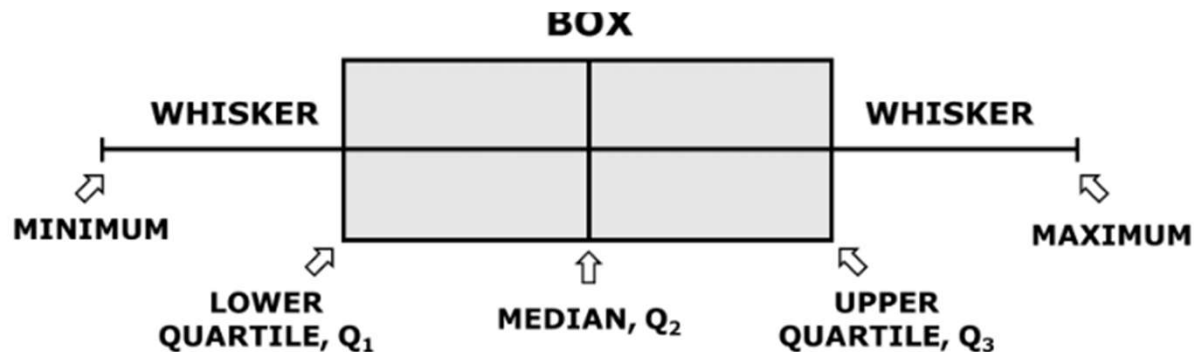
Range Plot

- Range Plot (also called Range Chart) uses a categorical attribute and a quantitative attribute, requiring more than one quantity per category.
- It shows the range (minimum and maximum) of the values.
- Often used to display the variability in data.



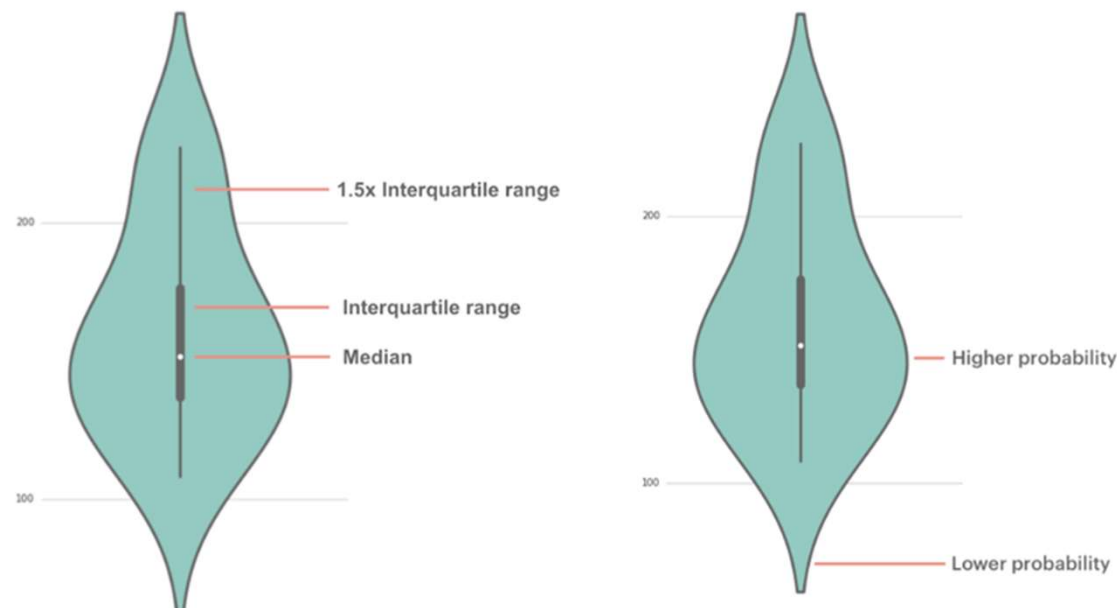
Box Plot

- A box plot) is **method of displaying data**, invented by J. Tukey.
- A visual display that summarizes data using a 'box and whiskers' format to show:
 - The minimum and maximum values (ends of the whiskers)
 - Interquartile range (length of the box)
 - Median (line through the box)



Violin Plot

- Violin plots are similar to box plots, except that they also show the probability density of the data at different values.
- They show summary statistics such as mean/median and interquartile ranges like box plots
- Also show the full distribution of the data.
- If the data has more than one peak, a violin plot shows their positions and relative amplitude.



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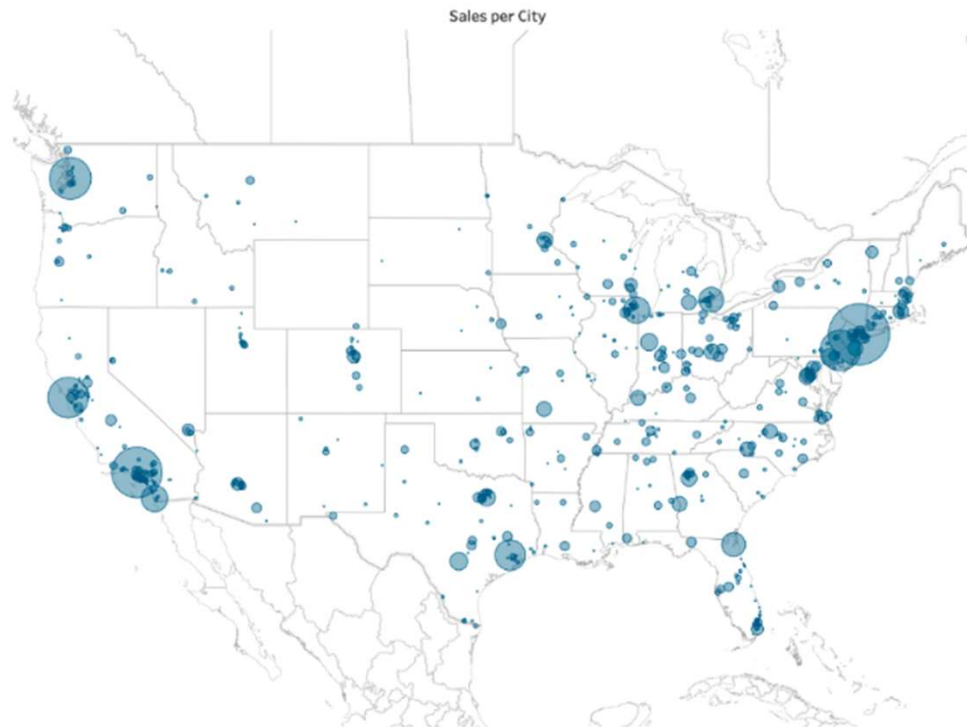
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Geographical Data

- Sometime the table may have data about various geographical locations.
- They are effectively visualized using maps showing different regions of the world.
- A cartographic projection maps longitude and latitude pairs to x, y coordinates.
- One can use projections to layout both geographic points (such as locations on a map) represented by longitude and latitude coordinates, or to project geographic regions (such as countries and states) .
- Various cartographic projections like *albers* are provided by the d3-geo library which can be used to create the visualizations.
- <https://github.com/d3/d3-geo#>

Symbol Map

- Symbol Map is a map that uses symbols (commonly circles) sized according to a quantitative variable.
- It combines geographic data from a geojson/json file with categorical and quantitative data from a table to show the magnitude of values in different regions.



Choropleth (Filled Map)

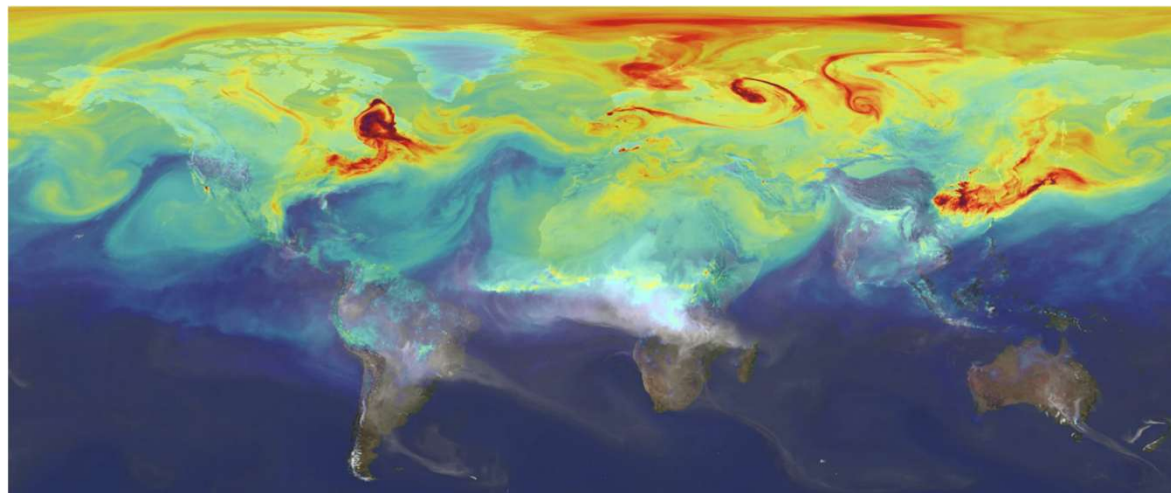
- Choropleth is a map that encodes quantitative values for geographic regions using color gradients.
- It requires a geojson/json file for geometry and another file with data in regions and a quantitative value, making it ideal for showing regional patterns and distributions.



Sales per state

Spatial Heatmap

- A spatial heat map displays the magnitude of a spatial attribute as color, usually cast over a map.
- Each square in a Heatmap is assigned a color representation based on the value of the cell.
- The magnitude of the value in that space determines the placement of color.



Reading

- Chapter 4